MiniOS Ekoms

User Manual

Operating Systems Final Project

Group 8 - SC1063

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1. Introduction

MiniOS Ekoms is a simple command-line operating system simulator written in Python. It runs inside Windows Command Prompt (CMD) and imitates basic operating system functions: a command shell, a file system, process management, memory management, user login, and system information. It is a simulation only. It is not a real operating system and does not change real Windows files, processes, or memory.

2. System Requirements

• Windows 10 or 11 with Command Prompt (CMD)

• Python 3.x installed and added to PATH

• No extra libraries or internet connection needed

3. Installation

• Download or clone the project from GitHub.

• Extract the folder if it is a ZIP file.

• Open CMD and go to the project folder, for example: `cd C:\Users\YourName\MiniOS`

4. How to Start MiniOS

In CMD, inside the project folder, type:

```
python main.py
```

The MiniOS banner appears, followed by the login screen.

5. Login Instructions

Type your username and press Enter, then type your password and press Enter. The password is shown as asterisks (*). You have 3 attempts. After 3 failed attempts, MiniOS locks and closes.

Username	Password	Role
admin	admin123	Administrator
student	student123	Standard User

After a successful login you will see the MiniOS prompt:

```
MiniOS>
```

6. Command List

Type a command at the `MiniOS>` prompt and press Enter. Commands are not case sensitive. Text in < > is something you must supply.

COMMAND: help

Purpose: Shows all available commands.

Syntax: `help`

Example: `help`

COMMAND: clear

Purpose: Clears the screen.

Syntax: `clear`

Example: `clear`

COMMAND: date

Purpose: Shows today's date.

Syntax: `date`

Example: `date`

COMMAND: time

Purpose: Shows the current time.

Syntax: `time`

Example: `time`

COMMAND: whoami

Purpose: Shows the user who is logged in.

Syntax: `whoami`

Example: `whoami`

COMMAND: systeminfo

Purpose: Shows simulated system information (OS name, version, CPU, memory, user).

Syntax: `systeminfo`

Example: `systeminfo`

COMMAND: version

Purpose: Shows the MiniOS version.

Syntax: `version`

Example: `version`

COMMAND: echo

Purpose: Prints the text you type.

Syntax: `echo <text>`

Example: `echo Hello MiniOS`

COMMAND: history

Purpose: Lists the commands you typed in this session.

Syntax: `history`

Example: `history`

COMMAND: users

Purpose: Lists the user accounts and their roles.

Syntax: `users`

Example: `users`

COMMAND: logout

Purpose: Logs out the current user and returns to the login screen.

Syntax: `logout`

Example: `logout`

COMMAND: exit

Purpose: Shuts down MiniOS. (shutdown does the same.)

Syntax: `exit`

Example: `exit`

COMMAND: dir

Purpose: Lists files and folders in the current directory.

Syntax: `dir`

Example: `dir`

COMMAND: mkdir

Purpose: Creates a new directory.

Syntax: `mkdir <folder>`

Example: `mkdir Documents`

COMMAND: create

Purpose: Creates a new empty file.

Syntax: `create <file>`

Example: `create notes.txt`

COMMAND: read

Purpose: Displays the contents of a file.

Syntax: `read <file>`

Example: `read notes.txt`

COMMAND: write

Purpose: Adds text to an existing file.

Syntax: `write <file> <text>`

Example: `write notes.txt Hello World`

COMMAND: delete

Purpose: Deletes a file or an empty directory.

Syntax: `delete <name>`

Example: `delete notes.txt`

COMMAND: rename

Purpose: Renames a file or directory.

Syntax: `rename <old> <new>`

Example: `rename notes.txt todo.txt`

COMMAND: cd

Purpose: Changes the current directory. Use .. to go up one level.

Syntax: `cd <folder>`

Example: `cd Documents`

COMMAND: pwd

Purpose: Shows the current directory path.

Syntax: `pwd`

Example: `pwd`

COMMAND: run

Purpose: Starts a simulated process and gives it a PID.

Syntax: `run <program>`

Example: `run calculator`

COMMAND: process

Purpose: Lists all running processes (PID, name, status).

Syntax: `process`

Example: `process`

COMMAND: kill

Purpose: Terminates a process by its PID.

Syntax: `kill <pid>`

Example: `kill 1`

COMMAND: memory

Purpose: Shows total, used, and free memory.

Syntax: `memory`

Example: `memory`

COMMAND: allocate

Purpose: Reserves an amount of memory (in MB).

Syntax: `allocate <MB>`

Example: `allocate 200`

COMMAND: free

Purpose: Releases memory you allocated (in MB).

Syntax: `free <MB>`

Example: `free 200`

7. Common Errors

MiniOS does not crash on mistakes. It shows a message and returns to the prompt.

Message	Cause	What to do
Unknown command.	The command is misspelled or does not exist.	Type **help** to see valid commands.
Error: File does not exist.	The file name is wrong or you are in a different directory.	Use **dir** to check the name and **pwd** to see where you are.
Error: A file or directory with that name already exists.	You tried to create something with a name already in use.	Pick another name or delete the old one first.
Error: Directory is not empty.	You tried to delete a folder that still has items.	Delete the items inside first.
Error: Process does not exist.	The PID is not in the process list.	Use **process** to see valid PIDs.
Error: PID must be a number.	You typed letters instead of a PID number.	Use a number, e.g. **kill 1**.
Error: Not enough memory.	The request is bigger than the free memory.	Use **memory** to check free memory, then allocate less.
Error: Cannot free ... MB.	You tried to release more than was allocated.	Free an amount no larger than what you allocated.
Error: Amount must be a whole number / greater than 0.	The memory amount was not a positive whole number.	Type a number such as **allocate 100**.
Error: Missing argument. Usage: ...	The command needs more information.	Add the missing part, following the usage shown.
Invalid username or password.	Wrong login details. After 3 failed tries the system locks.	Check your username and password; restart MiniOS if locked.

8. How to Exit MiniOS

Type:

```
exit
```

MiniOS shows *Shutting down MiniOS... Goodbye!* and returns you to CMD. Your files are saved automatically in `minios_disk.json` and will still be there next time you start MiniOS. Running processes and allocated memory are reset on each start.